
CliniScan

Chest X-Ray Analysis System

Milestone 4 — UI Interface Development & Deployment

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1. Executive Summary

Milestone 4 represents the final stage of the CliniScan project, transitioning the AI models from experimental Jupyter Notebooks into a production-ready web application. The primary objective was to create an intuitive, "clinician-first" interface that allows for real-time inference, multi-modal visualization (Bounding Boxes + Grad-CAM), and automated report generation. The system is containerized via Docker and deployed on Hugging Face Spaces for public accessibility.

2. Technical Architecture (The Stack)

The application follows a decoupled client-server architecture designed for high-performance CPU inference:

- **Backend (FastAPI):** A high-performance Python framework chosen for its asynchronous capabilities. It handles model orchestration, image preprocessing via *Albumentations*, and result serialization.
- **Frontend (Vanilla JS/CSS3):** A lightweight, single-page application (SPA) built without heavy frameworks to ensure maximum loading speed. It utilizes a modern dark-themed UI with *Syne* and *DM Mono* typography.
- **AI Engine: * Classification:** EfficientNet-B0 (AdamW optimized) for 14-class multi-label detection.
 - **Localization:** YOLOv8s for precise bounding box coordinate mapping.
 - **Explainability:** Integrated Grad-CAM library for heat-map generation.
- **Deployment:** Dockerized environment hosted on **Hugging Face Spaces**, ensuring cross-platform compatibility.

3. Core Interface Features

Based on the `index.html` and `app.py` implementation, the UI includes:

3.1 Single Scan Analysis

- **Interactive Toggles:** Users can enable/disable "Bounding Boxes" and "Grad-CAM Heatmaps" to save processing time.
- **Confidence Threshold Slider:** A dynamic slider (0.1 to 0.9) allowing clinicians to adjust the sensitivity of the AI detections.
- **Multi-Tab Viewer:** A custom image viewer that lets users switch between the *Original X-ray*, *Detected Regions*, and *Attention Maps (Grad-CAM)* without reloading the page.
- **Clinical Descriptions:** Automated logic that maps AI class IDs to human-readable clinical summaries (e.g., explaining what "Cardiomegaly" or "Pleural Effusion" indicates).

3.2 Batch Processing & Reporting

- Bulk Upload: An asynchronous queue system that processes multiple X-rays sequentially, showing a live progress list.
- PDF Report Generation: Utilizing the ReportLab engine, the system generates a downloadable "CliniScan Diagnostic Report" containing patient metadata, AI findings, and visualized coordinates.

3.3 System Health Monitoring

Hardware Telemetry: A dedicated sidebar footer that communicates with the /health endpoint to show real-time CPU status and model loading confirmation (Classification Model  / YOLO ).

4. Deployment Workflow

The deployment was executed through the following pipeline:

1. **Environment Sync:** Consolidating model checkpoints (.pt files) into a dedicated milestone3_cls/ckpt and milestone3_det directory structure.
2. **Containerization:** Writing a Dockerfile to install dependencies (OpenCV, Torch, Ultralytics) and expose port 7860.
3. **Hugging Face Integration:** Configuring the Space as a "Docker SDK" project, allowing for persistent hosting and easy updates via Git.

5. Summary of Deliverables

Deliverable	Description
Web Interface	Fully responsive HTML5/CSS3 dashboard
Inference API	FastAPI backend with 6 endpoints (/predict, /analyze, etc.).
Visualizer	Integrated Grad-CAM and Bounding Box overlay system.
Live Space	Publicly accessible URL for remote diagnostic testing.